গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED
FIVE STORIED RESIDENTIAL BUILDING

AT PORSHI-104, KAWAPARA, SYLHET.

CLIENT

AMIR ALI

Verification Code: p4WRa38X
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would be necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-50ft	10
02	Disturbed sample	02	0-50 ft	10
03	Disturbed sample	03	0-50 ft	10

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

<u>Bearing capacity for deep foundation</u>

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10, and Table 11,** respectively.

Table 7: The allowable bearing capacity for the shallow foundation in Tsf (FS=2.5)

Bore Hole No.	**Depth**	**Design N**	**Clay**						**Sand**
			Continuous			**Square(Isolated)**			
	(ft)	**-**	**D/B=0.25**	**D/B=0.5**	**D/B=1**	**D/B=0.25**	**D/B=0.5**	**D/B=1**	
BH01	**2.5-5.0**	**20**	2.343	2.464	2.687	2.767	2.990	3.252	2.525
	5.0-10	**20**	2.356	2.478	2.701	2.783	3.006	3.270	2.539
BH02	**2.5-5.0**	**19**	2.227	2.342	2.554	2.630	2.842	3.091	2.400
	5.0-10	**18**	2.032	2.137	2.330	2.400	2.593	2.820	2.190
BH03	**2.5-5.0**	**14**	1.624	1.708	1.862	1.918	2.072	2.254	1.750
	5.0-10	**14**	1.672	1.758	1.917	1.975	2.133	2.321	1.802
BH04	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH05	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH06	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH07	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH08	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH09	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH10	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.074	0.096	0.096	-	-	-	-	-	-	-
10	0.168	0.176	0.096	-	-	-	-	-	-	-
15	0.224	0.192	0.168	-	-	-	-	-	-	-
20	0.261	0.156	0.080	-	-	-	-	-	-	-
25	0.196	0.138	0.294	-	-	-	-	-	-	-
30	0.348	0.241	0.388	-	-	-	-	-	-	-
35	0.218	0.174	0.261	-	-	-	-	-	-	-
40	0.239	0.261	0.246	-	-	-	-	-	-	-
45	0.176	0.184	0.152	-	-	-	-	-	-	-
50	0.200	0.200	0.200	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.675	0.900	0.900	-	-	-	-	-	-	-
10	33.600	35.200	19.200	-	-	-	-	-	-	-
15	44.800	38.400	33.600	-	-	-	-	-	-	-
20	2.700	1.575	16.000	-	-	-	-	-	-	-
25	2.025	1.350	4.950	-	-	-	-	-	-	-
30	5.850	4.050	6.525	-	-	-	-	-	-	-
35	2.250	1.800	2.700	-	-	-	-	-	-	-
40	2.475	2.700	3.375	-	-	-	-	-	-	-
45	35.200	36.800	30.400	-	-	-	-	-	-	-
50	40.000	40.000	40.000	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.048	0.062	0.062	-	-	-	-	-	-	-
10	0.109	0.114	0.062	-	-	-	-	-	-	-
15	0.146	0.125	0.109	-	-	-	-	-	-	-
20	0.170	0.101	0.052	-	-	-	-	-	-	-
25	0.127	0.090	0.191	-	-	-	-	-	-	-
30	0.226	0.156	0.252	-	-	-	-	-	-	-
35	0.141	0.113	0.170	-	-	-	-	-	-	-
40	0.156	0.170	0.160	-	-	-	-	-	-	-
45	0.114	0.120	0.099	-	-	-	-	-	-	-
50	0.130	0.130	0.130	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=3.0)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.304	0.405	0.405	-	-	-	-	-	-	-
10	15.120	15.840	8.640	-	-	-	-	-	-	-
15	20.160	17.280	15.120	-	-	-	-	-	-	-
20	1.215	0.709	7.200	-	-	-	-	-	-	-
25	0.911	0.608	2.228	-	-	-	-	-	-	-
30	2.633	1.823	2.936	-	-	-	-	-	-	-
35	1.013	0.810	1.215	-	-	-	-	-	-	-
40	1.114	1.215	1.519	-	-	-	-	-	-	-
45	15.840	16.560	13.680	-	-	-	-	-	-	-
50	18.000	18.000	18.000	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed five-storied residential building.

Soil profile scenario

Overall soil information of the investigated site is not regular between the boreholes, and vertical inconsistency exists in the soil formation. The subsoil in borehole 1 is composed of soft to stiff clay, fine sand, and black organic. Subsoils in boreholes 2 and 3 are composed of soft to stiff clay and fine sand. At a greater depth than 45 feet from EGL, the examined soil in all boreholes indicated good bearing capability with a fine sand layer.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since clay layers are found in the top 40ft depth in all the boreholes, and black organic soil exists in borehole 1, a deep foundation (Bored/precast pile) is recommended to ensure the structure's stability and stability overcome the ground improvement complication. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found in the attached table (table 8-11) in the report based on the existing soil condition.

Safety provisions

Since the proposed site comprises soft, cohesive, black organic, and granular soil layers, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **site's settlement** considering the building load while designing deep foundations. Obviously, a pile load test covering the entire project area must be carried out to confirm the service pile's design parameters (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area and the quality management of the construction work, including workmanship, should be considered for the safe design of a shallow or deep foundation.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. The client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative, and the consultant or Design Engineer would decide the design engineer's selection of the type, depth, and dimensions of the foundation. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **50** ft, as the client provided.

Test Assisted By	**Report Prepared By**	**Countersigned By**
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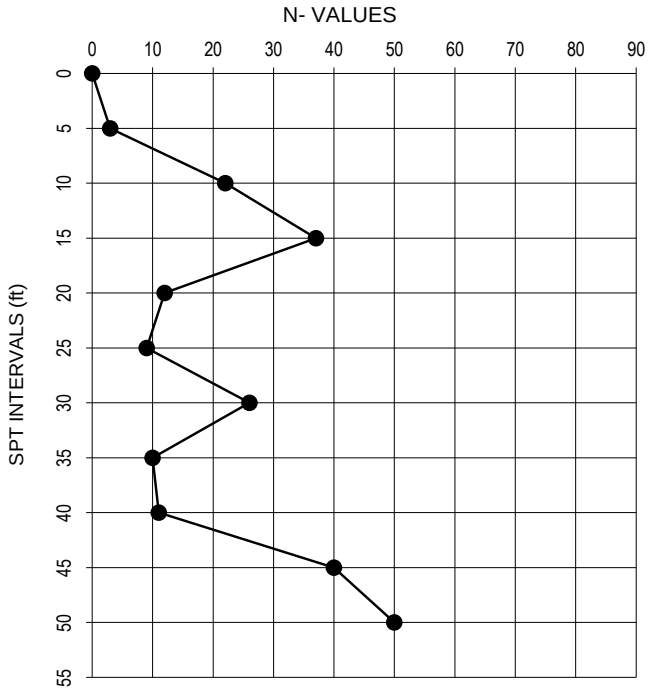
RECORD OF BORING AND TESTING

CLIENT : **AMIR ALI**	LOCATION: Poeshi-104, Kawapara, Sylhet.		Sandy clay with silt.
PROJECT NAME : **CONST. OF 05 STORIED RESIDENTIAL BUILDING .**	GROUND WATER LEVEL : 4'-0" BELOW EGL.		
BORE HOLE NO. **01**			Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 19/04/2022		
BORING DIA : **0.33ft**			Sand with silt & clay.
DEPTH OF BORING : 50 ft	COMPLETION DATE : 19/04/2022		
			Organic/Gravel/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value
						0	0	0	0	0
D1		5.0	5.0	Grey soft clay		5.0	1	1	2	3
D2		10.0	10.0	Grey fine sand		10.0	6	10	12	22
D3		15.0				15.0	10	15	22	37
D4		20.0	10.0	Black organic		20.0	2	3	9	12
D5		25.0				25.0	2	3	6	9
D6		30.0	10.0	Grey fine sand with silt & clay		30.0	3	10	16	26
D7		35.0				35.0	3	5	5	10
D8		40.0	5.0	Grey stiff clay		40.0	3	5	6	11
D9		45.0	10.0	Grey fine sand		45.0	10	18	22	40
D10		50.0				50.0	13	23	27	50

GRAPHICAL REPRESANTATION OF N- VALUES

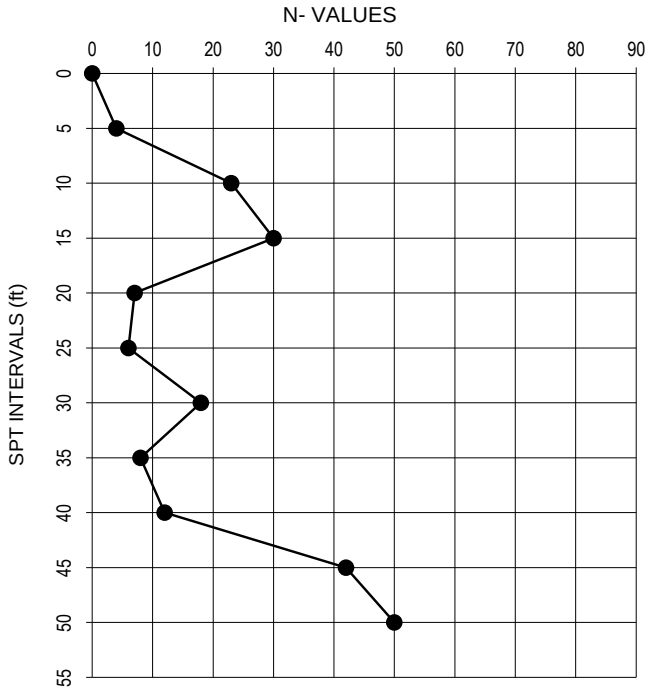


RECORD OF BORING AND TESTING

CLIENT : **AMIR ALI**	LOCATION: Poeshi-104, Kawapara, Sylhet.		Sandy clay with silt.
PROJECT NAME : **CONST. OF 05 STORIED RESIDENTIAL BUILDING .**	GROUND WATER LEVEL : 4'-0" BELOW EGL.		
BORE HOLE NO. **02**			Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 19/04/2022		
BORING DIA : **0.33ft**			Sand with silt & clay.
DEPTH OF BORING : 50 ft	COMPLETION DATE : 19/04/2022		
			Organic/Gravel/coarse sand

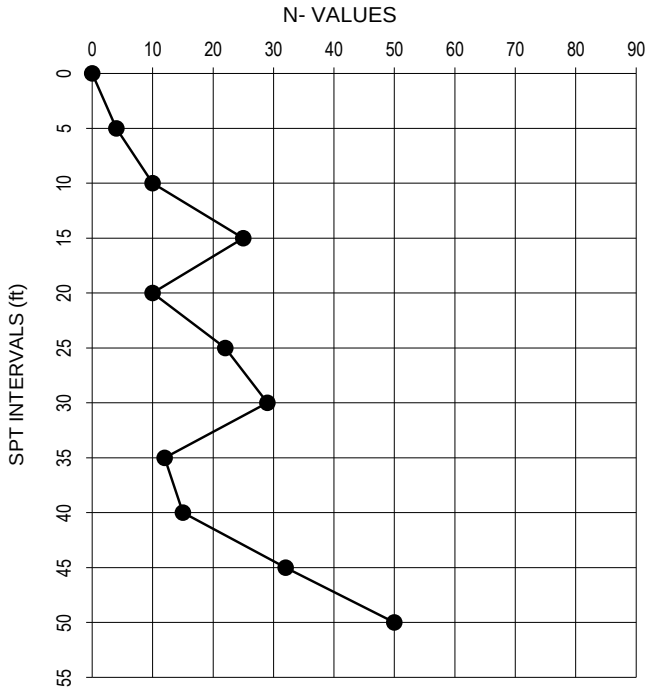
Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value
						0	0	0	0	0
D1		5.0	5.0	Grey soft clay		5.0	1	2	2	4
D2		10.0	10.0	Grey fine sand		10.0	7	11	12	23
D3		15.0				15.0	10	13	17	30
D4		20.0	10.0	Grey sandy clay		20.0	2	3	4	7
D5		25.0				25.0	3	3	3	6
D6		30.0	10.0	Grey stiff clay		30.0	4	7	11	18
D7		35.0				35.0	3	4	4	8
D8		40.0	5.0	Grey fine sand with silt & clay		40.0	4	6	6	12
D9		45.0	10.0	Grey fine sand		45.0	10	19	23	42
D10		50.0				50.0	13	23	27	50

GRAPHICAL REPRESANTATION OF N- VALUES



RECORD OF BORING AND TESTING

CLIENT : **AMIR ALI**	LOCATION: Poeshi-104, Kawapara, Sylhet.		Sandy clay with silt.
PROJECT NAME : **CONST. OF 05 STORIED RESIDENTIAL BUILDING .**	GROUND WATER LEVEL : 3'-0" BELOW EGL.		
BORE HOLE NO. **03**			Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 19/04/2022		
BORING DIA : **0.33ft**			Sand with silt & clay.
DEPTH OF BORING : 50 ft	COMPLETION DATE : 19/04/2022		
			Organic/Gravel/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value
						0	0	0	0	0
D1		5.0	5.0	Grey soft clay		5.0	1	2	2	4
D2		10.0	15.0	Grey fine sand with silt & clay		10.0	3	4	6	10
D3		15.0				15.0	10	12	13	25
D4		20.0				20.0	2	5	5	10
D5		25.0	10.0	Grey stiff clay		25.0	6	10	12	22
D6		30.0				30.0	10	13	16	29
D7		35.0	10.0	Grey fine sand with silt & clay		35.0	4	5	7	12
D8		40.0				40.0	5	7	8	15
D9		45.0	10.0	Grey fine sand		45.0	10	13	19	32
D10		50.0				50.0	14	24	26	50

GRAPHICAL REPRESANTATION OF N- VALUES